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Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

- a. Merge Sort
- b. Insertion Sort
- c. Quick Sort
- d. Bubble Sort

The correct answer is: Merge Sort

Question **2**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Arranging data in a specific order
- b. Finding a specific element in a list
- c. Removing duplicates from a list
- d. Merging two datasets

The correct answer is: Arranging data in a specific order

Question **3**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Sorting data in a single pass
b. Dividing the input into parts, solving each part, and combining the solutions
c. Rearranging data without sorting
d. Sorting data sequentially

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **4**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

a. sorted()
b. arrange()
c. order()
d. sort()

The correct answer is: sorted()

Question **5**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

a. order()
b. arrange()
c. sort()
d. sorted()

The correct answer is: sort()

Question **6**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Binary Search
b. Linear Search
c. Bubble Sort
d. Merge Sort

The correct answer is: Bubble Sort

Question **7**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Makes code execution slower
b. Provides a basis for other algorithms to work efficiently
c. Makes data harder to manage
d. Decreases the efficiency of algorithms

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **8**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Extraction
b. Selection
c. Distribution
d. Insertion

The correct answer is:
Insertion

Question **9**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It has no effect on the search operation
b. It increases the number of comparisons needed
c. It allows for faster searching
d. It makes the search operation slower

The correct answer is: It allows for faster searching

Question **10**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It sorts data out of the box efficiently
b. It requires external libraries
c. It is less efficient than custom sorting algorithms
d. It only works with integer arrays

The correct answer is: It sorts data out of the box efficiently

Question **11**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It bubbles up the largest element in each pass
b. It uses the heap data structure
c. It recursively sorts subproblems
d. It divides the list into sublists

The correct answer is: It bubbles up the largest element in each pass

Question **12**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

- a. Bubbling up the largest element to its correct position
- b. Sorting the entire list in one pass
- c. Bubbling up the smallest element
- d. Dividing the list into halves

The correct answer is: Bubbling up the largest element to its correct position

Question **13**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

- a. Bubble Sort cannot handle duplicate elements
- b. Bubble Sort is difficult to implement
- c. Bubble Sort is less efficient for large lists
- d. Bubble Sort does not guarantee sorted order

The correct answer is: Bubble Sort is less efficient for large lists

Question **14**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. Duplicates
- b. Selection
- c. All of the above
- d. Searching

The correct answer is: All of the above

Question **15**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Merge Sort
- b. Binary Search
- c. Linear Search
- d. Bubble Sort

The correct answer is: Merge Sort